FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Malnutrition & Growth Monitoring

Tutorial | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Namugga

Severe acute malnutrition (MUAC < 115 mm, weight-for-height Z-score below −3, or oedema of both feet) is a medical emergency in which the usual rules for fluids, feeding and infection are reversed; uncomplicated cases with a good appetite are treated at home with RUTF, while complicated cases need the WHO 10 steps, starting with hypoglycaemia, hypothermia, dehydration and infection and cautious F-75 feeding.

Must know before the ward round

1. Z-score cut-offs (WHO 2006 standards): wasting = weight-for-height/length (WHZ) < −2 (severe < −3); stunting = height-for-age < −2 (severe < −3); underweight = weight-for-age < −2 (severe < −3); overweight WHZ > +2, obesity > +3.
2. SAM in 6–59 months: WHZ < −3 OR MUAC < 115 mm OR bilateral pitting oedema. MAM: WHZ −3 to < −2 OR MUAC 115 to < 125 mm, no oedema.
3. Complicated SAM (admit): any IMCI danger sign or medical complication, oedema +++, or failed appetite test (cannot eat ⅓ of a 92 g RUTF sachet in 30 min). Otherwise outpatient therapeutic care with RUTF.
4. The 10 steps: hypoglycaemia, hypothermia, dehydration, electrolytes, infection, micronutrients, cautious feeding, catch-up growth, sensory stimulation, prepare for follow-up.
5. Hypoglycaemia < 3 mmol/L: 50 mL of 10% glucose or sugar water orally/NG; unconscious → 10% glucose 5 mL/kg IV; then feed F-75 every 30 min for 2 h.
6. F-75 (75 kcal and 0.9 g protein per 100 mL): 130 mL/kg/day (100 mL/kg/day if severe oedema), 2-hourly at 11 mL/kg/feed on days 1–2, 3-hourly at 16 mL/kg on days 3–5, 4-hourly at 22 mL/kg from day 6.
7. Transition: same volume of F-100 (or RUTF) for 2 days, then increase each feed by 10 mL until some is left (about 200 mL/kg/day; 150–220 kcal/kg/day). RUTF sachet 92 g = 500 kcal.
8. Antibiotics for ALL SAM. Complicated: ampicillin 50 mg/kg IV/IM 6-hourly for 2 days then amoxicillin for 5 days, PLUS gentamicin 7.5 mg/kg once daily for 7 days. Uncomplicated (UCG 2023): amoxicillin 40 mg/kg twice daily for 5 days.
9. No IV fluids unless shocked (15 mL/kg over 1 h); dehydration = ReSoMal 5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h. No diuretics for oedema. No iron until rehabilitation (never with RUTF).
10. Weight gain in rehabilitation: good > 10 g/kg/day, moderate 5–10, poor < 5 (full reassessment).
11. Exit from treatment: UCG 2023 = WHZ ≥ −2 or MUAC ≥ 125 mm (same indicator as admission) and no oedema for 2 weeks. WHO 2023 = WHZ ≥ −2 AND MUAC ≥ 125 mm on 2 consecutive visits and no oedema.
12. Uganda (UDHS 2022): 26% of under-fives stunted, 2.9% wasted, 10.2% underweight; Karamoja is worst affected.

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In a nutshell

Malnutrition here means **undernutrition**: a child who is too thin (**wasting**), too short (**stunting**), swollen with **nutritional oedema** (kwashiorkor), or deficient in micronutrients. It underlies **about half of all child deaths** (WHO), because malnourished children die of ordinary infections: pneumonia, diarrhoea, malaria, measles, TB and HIV.

Growth monitoring is how you catch it early: weigh and measure every child, plot on the **WHO growth chart**, and act when the line **flattens or falls** rather than waiting for the child to look wasted.

Severe acute malnutrition (SAM) is a medical emergency with its own rulebook. The physiology is "turned down" (reductive adaptation), so the child **cannot handle** normal amounts of fluid, sodium, protein or iron. Treat in the WHO order: **hypoglycaemia, hypothermia, dehydration, electrolytes, infection**, then **micronutrients and cautious feeding**, then **catch-up growth, stimulation and follow-up**.

> **CLINICAL PEARL**
> The one thing never to forget: **a child with SAM can look "not that sick" and die overnight** of hypoglycaemia, hypothermia, hidden sepsis, or heart failure from your IV drip. Feed every 2 hours **including at night**, keep them warm, give antibiotics to all, and never run IV fluids unless the child is in shock.

Definitions & classification

Anthropometric indices (WHO Child Growth Standards 2006, 0–5 years)

Index	Reflects	Moderate	Severe
Weight-for-length/height (WLZ/WHZ)	**Wasting** (acute)	**< −2 to −3**	**< −3**
Length/height-for-age (LAZ/HAZ)	**Stunting** (chronic)	**< −2**	**< −3**
Weight-for-age (WAZ)	**Underweight** (mixed; used on the child health card)	**< −2**	**< −3**
MUAC (6–59 months)	Wasting and mortality risk	**115 to < 125 mm**	**< 115 mm**
BMI-for-age (5–19 years, WHO 2007)	Thinness / overweight	Thinness < −2; overweight > +1	Severe thinness < −3; obesity > +2
Overweight / obesity (under 5)	Overnutrition	WHZ **> +2** (overweight)	WHZ **> +3** (obesity)

Z-score = (child's measurement − median of the reference population) ÷ standard deviation. A Z-score of −2 means 2 SD below the median (about the 2nd–3rd centile).

Acute malnutrition (WHO 2013/2023, UCG 2023)

Category	Criteria (6–59 months)	Where to treat
MAM (moderate wasting)	WHZ −3 to < −2 **or** MUAC 115 to < 125 mm, **no oedema**	Counselling, family foods, **supplementary food** (LNS/RUSF) where available
SAM, uncomplicated	WHZ < −3 **or** MUAC < 115 mm **or** oedema + or ++, **and** passes the appetite test, **no** danger signs or medical complications	**Outpatient therapeutic care (OTC)** with RUTF
SAM, complicated	SAM **plus** any: IMCI danger sign, medical complication (pneumonia, diarrhoea with dehydration, hypoglycaemia, hypothermia, severe anaemia, fever, skin breakdown), **oedema +++**, **failed appetite test**, or age < 6 months	**Inpatient therapeutic care (ITC)** / nutrition unit

Grading nutritional oedema (press both feet for 3 seconds)

Grade	Extent
+	Both feet
++	Feet plus lower legs, hands or lower arms
+++	Generalised, **including the face**

Infants under 6 months (WHO 2023 "infants at risk of poor growth and development")

WHO 2023 no longer uses "SAM" alone for young infants. Infants < 6 months are **at risk** if they have, for example, **WAZ < −2**, **WLZ < −3**, **MUAC < 110 mm (6 weeks to < 6 months)**, oedema, **failure to gain weight on two consecutive measurements**, or **ineffective breastfeeding**. Admit if there are danger signs, acute medical problems, oedema or **recent weight loss**.

Clinical syndromes (old names you will still hear)

Syndrome	Features
Marasmus (severe wasting)	"Skin and bones", **old man's face**, loose skin folds on the buttocks ("baggy pants"), prominent ribs and scapulae, **alert, hungry**, no oedema
Kwashiorkor (nutritional oedema)	**Bilateral pitting oedema**, moon face, **apathy and misery**, **flaky-paint dermatosis**, skin peeling and ulceration, **sparse, pale, straight, easily plucked hair** (flag sign), fatty **hepatomegaly**, poor appetite; weight may look normal because of fluid
Marasmic kwashiorkor	Wasting **and** oedema: the highest mortality

Epidemiology & Uganda context

- **Globally (2024)**: about **150 million** under-fives stunted, **43 million** wasted and **36 million** overweight; malnutrition is associated with **about half of all child deaths** (WHO IYCF fact sheet).
- **Uganda (UDHS 2022)**: **26% of under-fives are stunted** (down from 33% in 2011 and 29% in 2016), **2.9% wasted** and **10.2% underweight**. Stunting is highest in **Karamoja, West Nile, Acholi** and refugee-hosting districts; wasting is concentrated in **Karamoja**. About **half of under-fives are anaemic** (UDHS 2016).
- **Overweight** is rising (a "triple burden": undernutrition, overnutrition, micronutrient deficiency).
- **Drivers in Uganda**: suboptimal **exclusive breastfeeding** and **complementary feeding** (only about **15% of children 6–23 months get a minimum acceptable diet**, UDHS 2016), repeated **diarrhoea, malaria and**

worms, **HIV**, poverty and food insecurity, short birth intervals, teenage pregnancy, and **low birth weight** (about 1 in 10 babies).

- **Services**: Uganda follows **Integrated Management of Acute Malnutrition (IMAM, MoH 2016)**: community screening with **MUAC tapes** by VHTs, **OTC** at health centres, **ITC** at hospitals (e.g. **Mwanamugimu Nutrition Unit, Mulago**), and supplementary feeding in emergency-affected districts. In FY 2022/23 only about **45% of health facilities** provided IMAM services.

> UGANDA CONTEXT
> **What is actually available**: **MUAC tapes** and scales at most HC IIIs; **RUTF** (Plumpy'Nut, 92 g sachets) at OTC sites, but **stock-outs** occur; **F-75 and F-100** (sachets or locally prepared from dried skimmed milk, sugar, oil and the electrolyte/mineral mix) mainly at hospitals; **ReSoMal** sachets at nutrition units. **Weight-for-length charts** are often missing at HC level, so MUAC and oedema are the practical admission criteria. Inpatient case fatality on Ugandan nutrition units remains high, especially in children with **HIV**, so the 10 steps must be followed exactly.

Pathophysiology

Reductive adaptation

In chronic starvation the body **slows down** to survive: lower metabolic rate, less heat production (→ **hypothermia**), reduced liver gluconeogenesis and glycogen (→ **hypoglycaemia**), thin weak heart muscle (→ **heart failure** with modest fluid), reduced kidney ability to excrete sodium and water, a "leaky" **sodium–potassium pump** (cells lose **potassium and magnesium** and gain **sodium**), atrophic gut and pancreas (→ **malabsorption, diarrhoea**), and a suppressed immune system (→ **infection without fever**). Treatment must work **with** this slowed state: small, frequent, low-protein, low-sodium feeds first; rebuild later.

Why give these specific treatments

Problem	Mechanism	Hence
Hypoglycaemia	Low glycogen; infection	Feed **every 2–3 h day and night**; 10% glucose
Hypothermia	Low metabolic rate, little fat	Warm, cover the head, kangaroo care
Oedema	Sodium retention, potassium loss, oxidative stress, low albumin	**No diuretics**; low salt; extra K and Mg
Infection without fever	Immune suppression	**Antibiotics for every child**
Heart failure	Thin myocardium, sodium load	**No IV fluid** unless shock; slow ReSoMal; F-75
Iron toxicity	Iron feeds bacteria and free radicals; low transferrin	**No iron** in stabilisation
Refeeding syndrome	Insulin surge drives phosphate, K and Mg into cells	Start with **F-75**, increase slowly

Kwashiorkor: why oedema?

The cause is not simply "low protein". Contributing ideas: **oxidative stress** (free radicals outstrip depleted antioxidants such as glutathione, zinc and selenium), **sodium retention and potassium depletion**, **low albumin**, **aflatoxins**, and **gut microbiome** changes. The practical lesson: oedematous children are **sicker**, **less hungry** and **more fragile** than they look.

Growth physiology (normal benchmarks)

Parameter	Benchmark
Birth weight	Regained by **10–14 days**; **doubles by about 4–5 months**, **triples by 12 months**, quadruples by about 2 years
Weight gain	About **25–30 g/day** in the first 3 months, falling to about 10 g/day by 12 months
Length	About **50 cm at birth**, **75 cm at 1 year**, about **100 cm at 4 years**
Head circumference	About **35 cm at birth**, about **47 cm at 1 year**
MUAC	Fairly constant (about 13–16 cm) between **6 and 59 months**, which is why one cut-off works across that age range

Stunting

Stunting reflects **chronic** under-nutrition and repeated infection from **conception to 2 years** (the "first 1000 days"). It is largely **irreversible** after age 2, and is linked to poorer school performance and adult earnings and, when followed by rapid weight gain, to adult obesity and diabetes.

Clinical acumen

Focused history

Ask	Why it matters
Feeding history: breastfeeding (exclusive? still?), age at complementary foods, what, how thick, how often, who feeds, own plate?	Most malnutrition in Uganda is a feeding problem; tailors counselling
24-hour dietary recall	Energy density and diversity (animal-source foods, oil, vitamin A-rich foods)
Appetite now	Poor appetite = complicated SAM or infection
Weight trajectory on the child health card	Growth faltering, when it started
Diarrhoea, vomiting, cough > 2 weeks, fever, night sweats	Complications; **TB**
Swelling of feet and face: when it started	Nutritional oedema vs nephrotic, heart or liver disease
HIV status of mother and child; EID; ART; co-trimoxazole	HIV is a major driver and changes treatment
TB contacts, measles exposure	TB screening; measles worsens malnutrition
Immunisation (UNEPI card), **vitamin A** and **deworming** history	Catch-up; measles vaccine on admission
Birth history: weight, prematurity, twins	Low birth weight and catch-up
Family and social: mother's health and mood, father present, siblings' nutrition, food security, income, distance to clinic	Risk of relapse; social support; depression in mother
Previous episodes of SAM or admissions	Relapse risk; deeper causes (HIV, cerebral palsy, heart disease)
Development	Malnutrition delays development; cerebral palsy causes feeding difficulty

Examination

1. **Danger signs and ETAT**: unable to drink/breastfeed, vomits everything, convulsions, lethargy/unconsciousness, shock, severe respiratory distress.
2. **Anthropometry**: **weight** (undressed, calibrated scale), **length** (lying board under 2 years or < 87 cm) or **height**, **MUAC** (left arm, midpoint between acromion and olecranon, arm hanging relaxed, tape snug not tight), **head circumference**; plot **WHZ, HAZ, WAZ**.
3. **Oedema**: press **both feet** with the thumbs for 3 seconds; grade +/++/+++.
4. **Temperature** (axillary; rectal if hypothermia suspected), **RBS**.
5. **Hydration** (unreliable in SAM), **circulation** (cold hands, CRT, pulse).
6. **Eyes**: **Bitot's spots**, conjunctival and corneal **xerosis**, corneal **ulceration/clouding** (keratomalacia: emergency vitamin A), photophobia.
7. **Mouth**: angular stomatitis, cheilosis, **oral thrush** (HIV), ulcers.
8. **Skin and hair**: flaky-paint dermatosis, ulceration, peeling, depigmentation, hair changes, scabies, infected sores.
9. **Signs of infection** (often hidden): fast breathing, chest signs, ear discharge, lymph nodes, hepatosplenomegaly.
10. **Signs of HIV**: lymphadenopathy, parotid enlargement, thrush, recurrent infections.
11. **Signs of heart failure**: big tender liver, gallop, crackles, raised JVP.
12. **Development and interaction**: apathy, irritability, milestones.

Red flags

> RED FLAG
>
> **Hypoglycaemia (RBS < 3 mmol/L), hypothermia (axillary < 35 °C / rectal < 35.5 °C), lethargy or unconsciousness, shock, corneal ulceration or clouding, oedema +++, severe anaemia (Hb < 4 g/dL), severe pneumonia, dehydration from profuse diarrhoea, failed appetite test, and any infant under 6 months with weight loss.** A falling temperature or glucose usually means **sepsis**. A rising pulse and breathing rate during feeding or rehydration means **heart failure**.

Thinking like a clinician

- **Measure every sick child.** Wasting is easy to miss in a crying toddler under a blanket, and oedema can make a malnourished child look "well fed".
- **The appetite test is a clinical test**, not a formality: a child who cannot eat RUTF is sick and needs to be in hospital.
- **Infection has no fever in SAM.** Hypothermia, hypoglycaemia, lethargy and refusal to feed are its signs.
- **"Sunken eyes" and "slow skin pinch" are features of wasting itself**; diagnose dehydration from the **history of watery stools** and recent change.
- **Oedema + proteinuria = nephrotic syndrome**, not kwashiorkor: always **dip the urine**. Also think **heart failure** and **liver disease**.
- **No improvement?** Look for **TB, HIV, UTI, otitis media, hidden pneumonia, giardia**, feeding errors (diluted F-75, missed night feeds) and maternal depression.
- **Common student mistakes**: putting the child on a drip; giving furosemide for oedema; giving iron syrup on admission; skipping night feeds; giving high-dose vitamin A to everyone; prescribing ORS 75 mL/kg; forgetting the measles vaccine; using percentage weight gain as the discharge criterion.

Clinical workup

Step 0: First 5 minutes

1. **Triage (ETAT)**: emergency signs? Treat shock (**15 mL/kg over 1 h**, glucose-containing fluid), convulsions, airway.
2. **Measure**: MUAC, oedema, weight; classify SAM.
3. **RBS**: **< 3 mmol/L → 50 mL of 10% glucose or sugar water** (1 rounded teaspoon of sugar in 3 tablespoons of water) orally/NG; **unconscious → 10% glucose 5 mL/kg IV**; then F-75 straight away.
4. **Temperature**: hypothermic → **skin-to-skin with mother**, cover the head, blankets, warm room.
5. **Eyes**: corneal ulcer → vitamin A now, atropine and antibiotic drops, pad the eye.
6. **First antibiotic dose** and **first feed** within 30 minutes of arrival. At an HC: give sugar water, keep warm, give the first dose of ampicillin + gentamicin, and refer.

Step 1: Focused history

- [ ] **Feeding** (breastfeeding, complementary foods, 24-h recall) → counselling and cause.
- [ ] **Appetite** → complicated vs uncomplicated.
- [ ] **Diarrhoea/vomiting** → ReSoMal and complications.
- [ ] **Cough, fever, TB contact** → TB work-up.
- [ ] **HIV status** (child and mother) → ART, co-trimoxazole, prognosis.
- [ ] **Measles, immunisation, vitamin A, deworming** → catch-up.
- [ ] **Onset of oedema** → kwashiorkor vs other causes.
- [ ] **Social situation and maternal health** → discharge planning, relapse prevention.

Step 2: Focused examination

As above. **Document**: weight, length, **WHZ**, **MUAC**, **oedema grade**, temperature, RBS, Hb, eye signs, skin grade, signs of infection, signs of heart failure.

Document these positives	And these negatives
"MUAC 10.8 cm, WHZ < −3, oedema ++"	"No corneal lesions"
Temperature and RBS on arrival	"No signs of heart failure: liver 1 cm, no gallop"
Appetite test result (amount in 30 min)	"No dehydration history"
Skin lesions (site, extent)	"Urine: no protein" (excludes nephrotic)

Step 3: Bedside tests

Test	How	What it tells you
MUAC	Left mid-upper arm, colour-coded tape	**Red < 115 mm** (SAM), **yellow 115–124** (MAM), **green ≥ 125**
Weight and length → WHZ	WHO tables/charts by sex	SAM, MAM, normal; baseline for weight-gain monitoring
Appetite test	Offer RUTF in a quiet corner for 30 min with clean water	**Pass**: eats at least **⅓ of a 92 g sachet** (or 3 teaspoons from a pot). **Fail** → admit
RBS	Glucometer	Hypoglycaemia (< 3 mmol/L)
Temperature	Axillary (rectal if low)	Hypothermia < 35 °C axillary / < 35.5 °C rectal
HemoCue	Finger prick	Severe anaemia (Hb < 4, or 4–6 with distress)

Test	How	What it tells you
Malaria RDT	Finger prick	Treat malaria
HIV rapid test	With counselling	Start ART after stabilisation
Urine dipstick	Fresh urine	Protein (nephrotic), nitrites/leucocytes (UTI), glucose

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
FBC / Hb	All admissions	Anaemia (iron, folate, malaria, HIV); low WBC in sepsis	Transfusion only if Hb < 4 or 4–6 with distress; iron later. HC IV and above
Blood smear / RDT	Fever or not (fever often absent)	Malaria	Artesunate / ACT
HIV test (DNA PCR if < 18 months and exposed)	**Every** child with SAM	Positive	ART after stabilisation (about 1–2 weeks), co-trimoxazole
Electrolytes (Na, K, Mg, PO_4)	Complicated SAM, diarrhoea, oedema, during refeeding	**Low K, Mg, PO_4**; Na often low (dilutional; do not correct with saline)	Electrolyte mix in feeds; slow feeding if PO_4 falls. **RRH/NRH**
Blood culture	Sepsis, hypothermia, hypoglycaemia, no response at 48 h	Gram-negatives, *Salmonella*, *S. aureus*	Change to ceftriaxone or targeted therapy. RRH/NRH
Urinalysis and culture	All admissions	UTI is common and silent	Antibiotic choice
Stool microscopy	Diarrhoea, persistent diarrhoea	Giardia, ova, cysts	Metronidazole for giardia; deworm later
Chest X-ray	Cough, TB contact, no improvement	TB, pneumonia, cardiomegaly	TB treatment (Xpert on gastric aspirate/stool)
Xpert MTB/RIF Ultra (sputum, gastric aspirate, **stool**)	TB suspected	MTB detected	Start TB treatment; contact tracing
Albumin, LFTs	Oedema of unclear cause	Low albumin in kwashiorkor and nephrotic	Exclude nephrotic (proteinuria) and liver disease

Step 5: Putting it together

Model one-line summary

> "A 14-month-old boy, 6.2 kg, length 71 cm (WLZ < −3, LAZ −2.5), MUAC 10.9 cm, no oedema, with 5 days of cough and fast breathing, hypothermic (35.2 °C axillary) and hypoglycaemic (2.4 mmol/L), HIV-exposed with unknown status, missed measles vaccine: **complicated SAM (severe wasting) with severe pneumonia, hypoglycaemia and hypothermia**, and moderate stunting."

Problem list

1. Complicated SAM (severe wasting) → 10 steps, F-75.
2. Hypoglycaemia and hypothermia (probable sepsis).
3. Severe pneumonia (rule out TB).
4. HIV exposure, status unknown → DNA PCR.
5. Stunting and missed measles–rubella vaccine.

EXAM FAVOURITE

Worked example 1: writing the F-75 order. Weight 6.2 kg, no oedema → **130 mL/kg/day = 806 mL/day**.

- **Days 1–2: 2-hourly (12 feeds) × 11 mL/kg = 68 mL** per feed (≈ 70 mL).
- **Days 3–5: 3-hourly (8 feeds) × 16 mL/kg = 99 mL** per feed (≈ 100 mL).
- **Day 6 onwards: 4-hourly (6 feeds) × 22 mL/kg = 136 mL** per feed. If **severe oedema (+++)**: 100 mL/kg/day instead. **NG tube** if he takes **less than 80%** of the feed at **two consecutive feeds**. Breastfeed **before** each F-75 feed if still breastfeeding.

EXAM FAVOURITE

Worked example 2: weight gain during rehabilitation (UCG method). Weight today 6300 g; 3 days ago 6000 g. Gain = 300 g ÷ 3 days = 100 g/day. Average weight = (6000 + 6300) ÷ 2 = 6.15 kg. Rate = 100 ÷ 6.15 = **16.3 g/kg/day = good (> 10)**. If it were **5–10 g/kg/day** (moderate): check intake targets and look for missed infection. **< 5 g/kg/day** (poor): full reassessment (TB, HIV, UTI, feeding errors, mother's support).

EXAM FAVOURITE

Worked example 3: plotting and interpreting a growth chart. A girl's weight-for-age on her child health card follows the **0 Z-score line** to 6 months (exclusively breastfed), then **flattens** from 7 to 10 months and crosses below the **–2 line** at 11 months. Interpretation: **growth faltering after starting complementary feeding** (late or thin porridge, too few meals, repeated diarrhoea or malaria). Action **now**: measure MUAC and WLZ, take a feeding history, check for illness and HIV exposure, counsel on complementary feeding (thick porridge enriched with oil, groundnut paste, egg or fish; 3–4 meals and 1–2 snacks), and review in 2–4 weeks. Do not wait for her to become severely wasted.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Kwashiorkor (nutritional oedema)	Oedema starting in the feet, skin and hair changes, apathy, poor diet; urine protein negative	Clinical; urinalysis
Nephrotic syndrome	Periorbital oedema first, **heavy proteinuria (+++/++++)**, frothy urine, otherwise well nourished	Urine dipstick, albumin, cholesterol
Acute glomerulonephritis	Puffy face, **haematuria**, hypertension, oliguria	Urinalysis, BP, creatinine
Congestive heart failure (rheumatic, congenital, severe anaemia)	Murmur, gallop, big tender liver, dyspnoea; poor weight gain	Echo, CXR, Hb
Liver disease (cirrhosis, schistosomiasis)	Ascites, splenomegaly, jaundice	LFTs, ultrasound
HIV with wasting	Persistent diarrhoea, thrush, lymphadenopathy, recurrent infections	HIV test / DNA PCR
Tuberculosis	Cough > 2 weeks, contact, failure to gain weight, fever	CXR, Xpert, TST
Chronic diarrhoea / malabsorption	Persistent diarrhoea, bulky stools	Stool microscopy, HIV test
Cerebral palsy / feeding disability	Abnormal tone, delay, choking with feeds	Developmental exam
Congenital heart disease	Sweating and breathlessness with feeds, murmur, cyanosis	Echo

Condition	Distinguishing features	Key investigation
Endocrine and genetic short stature	**Stunted but well proportioned**, not wasted; dysmorphism; hypothyroid features	Thyroid function, karyotype (specialist)
Malignancy (Burkitt, leukaemia)	Mass, pallor, bleeding, organomegaly	FBC, ultrasound, biopsy

Investigations

Test	What you are looking for	Why
Weight, length, MUAC, oedema	Classification	Admission, treatment, exit criteria
RBS, temperature	Hypoglycaemia, hypothermia	Immediate treatment
Hb, malaria test	Anaemia, malaria	Transfusion and antimalarial
HIV test	HIV infection	ART, co-trimoxazole, prognosis
Urinalysis	Protein, UTI	Differential; antibiotics
CXR, Xpert, TST	TB	TB treatment
Electrolytes, phosphate	K, Mg, PO_4	Refeeding risk
Stool microscopy	Parasites	Targeted treatment

Management

A. Growth monitoring and promotion (prevention)

1. **Weigh every child under 2 years every month** (and at every clinic contact) and **plot on the child health card**; measure **length** at least every 3 months; **MUAC** from 6 months at every contact (VHT screening).
2. **Interpret the line, not the dot**: rising parallel to the curves = good; **flat** = growth faltering; **falling** = danger; crossing **two major lines** = serious.
3. **Act on faltering**: feeding history, illness check (malaria, diarrhoea, TB, HIV), counsel the caregiver (see the IYCF topic), deworm (from 1 year, 6-monthly), **vitamin A** (6-monthly from 6 months to 5 years), immunise, and **review in 2–4 weeks**.
4. Plot **head circumference** under 2 years and **BMI-for-age** in school-age children.

B. Moderate acute malnutrition (MAM)

1. Assess fully and **treat illnesses** (WHO 2023 B12): malaria, worms, TB, HIV.
2. **Counsel** on feeding for age: nutrient-dense family foods, **animal-source foods**, oil or groundnut paste added to porridge, extra meals.
3. **Specially formulated foods** (WHO 2023 B13–B16): prioritise **lipid-based nutrient supplements (LNS / RUSF)** for higher-risk children (MUAC 115–119 mm, WAZ < −3, age < 24 months, relapse, previous SAM, HIV/TB, disability, mother dead or unwell) and for **all** children in humanitarian settings; aim to supply **40–60%** of daily energy (total about 100–130 kcal/kg/day).
4. **Follow up**: UCG 2023 **every 30 days**; refer if worse or losing weight.

C. Uncomplicated SAM: outpatient therapeutic care (OTC)

1. **Admission criteria met**, **passes appetite test**, no danger signs, no complications.
2. **Amoxicillin 40 mg/kg twice daily for 5 days** (UCG 2023).
3. **RUTF** (UCG 2023 table; one 92 g sachet = **500 kcal**):

Weight (kg)	Rehabilitation: sachets/day	Sachets/week
4.0–4.9	2	14
5.0–6.9	2.5	18
7.0–8.4	3	21
8.5–9.4	3.5	25
9.5–10.4	4	28
10.5–11.9	4.5	32
≥ 12	5	35

*This provides about 200 kcal/kg/day. **WHO 2023** recommends **150–185 kcal/kg/day** until recovery, optionally reducing to **100–130 kcal/kg/day** once the child is no longer severely wasted and has no oedema.*

4. **How to give RUTF**: wash hands; small, regular meals **5–6 times a day**; **breastfeed first** before each RUTF meal; offer **plenty of clean water** from a cup; RUTF only for 2 weeks, then add family foods **after** the daily RUTF ration; RUTF is **medicine**, not to be shared.
5. **Malaria** test and treat; **measles–rubella vaccine** if due; **HIV test**; **TB screen**; **deworm** at the second visit (albendazole 200 mg if 12–23 months, 400 mg if ≥ 2 years).
6. **No extra iron, zinc, folate or vitamin A** (all in RUTF) unless specific signs.
7. **Follow up weekly or every 2 weeks** (UCG: 7 days, then every 14 days): weight, MUAC, oedema, appetite test, illness. **Refer** if appetite fails, danger signs, oedema worsens, weight loss for 2 visits or static for 3, or no response.

D. Complicated SAM: the WHO 10 steps (inpatient)

Step	Stabilisation: days 1–2	Days 3–7	Rehabilitation: weeks 2–6
1. Hypoglycaemia	✔		
2. Hypothermia	✔		
3. Dehydration	✔		
4. Electrolytes	✔	✔	✔
5. Infection	✔	✔	
6. Micronutrients	✔ **no iron**	✔ no iron	✔ **with iron** (if F-100)
7. Cautious feeding (F-75)	✔	✔	
8. Catch-up growth (F-100/RUTF)			✔
9. Sensory stimulation	✔	✔	✔
10. Prepare for follow-up			✔

Step 1: Hypoglycaemia (< 3 mmol/L)

- Conscious: **50 mL of 10% glucose or sugar water** orally or NG, then **F-75 every 30 min for 2 h** (a quarter of the 2-hourly feed each time), then 2-hourly day and night.
- Unconscious, lethargic or convulsing: **10% glucose 5 mL/kg IV**, then 50 mL by NG; if no IV, sugar under the tongue (1 teaspoon moistened) every 20 min.
- **Antibiotics** now. Recheck glucose at **30 min**.

Step 2: Hypothermia (axillary < 35 °C, rectal < 35.5 °C)

- **Feed** immediately; **skin-to-skin (kangaroo)** with the mother, cover with blankets, **hat**; warm room, no draughts; heater not pointed at the child; **no hot water bottles**.
- **Antibiotics**; check glucose. Temperature **2-hourly** (every 30 min if using a heater).

Step 3: Dehydration (see the Diarrhoea topic)

- **No IV unless shock. ReSoMal 5 mL/kg every 30 min for 2 h**, then **5–10 mL/kg/h for 4–10 h**, alternating with F-75; **ReSoMal 50–100 mL per watery stool** after that (UCG). If ReSoMal is not available: **low-osmolarity ORS** (WHO 2023 B7).
- **Shock**: **15 mL/kg over 1 h** of RL with 5% glucose (or half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose); repeat once if improving; if not, **whole blood 10 mL/kg over 3 h** and treat septic shock.
- **Stop** if pulse rises by 25 and RR by 5, or the liver enlarges.

Step 4: Electrolytes

- Extra **potassium 3–4 mmol/kg/day** and **magnesium 0.4–0.6 mmol/kg/day** (in F-75/F-100/RUTF and ReSoMal, or 20 mL of the electrolyte/mineral solution per litre of feed).
- **Low-salt** food; **no diuretics** for oedema. **Do not** treat low sodium with saline.

Step 5: Infection (antibiotics for every child)

Situation	Antibiotic	Dose	Route	Frequency	Duration
Uncomplicated SAM (OTC)	**Amoxicillin**	**40 mg/kg** (UCG 2023)	Oral	**12-hourly**	**5 days**
Complicated SAM	**Ampicillin** (or benzylpenicillin 50,000 IU/kg)	**50 mg/kg**	IV/IM	**6-hourly**	**2 days**, then
	→ **Amoxicillin**	**25–40 mg/kg**	Oral	**8-hourly**	**5 days**
	PLUS gentamicin	**7.5 mg/kg**	IV/IM	**Once daily**	**7 days**
No improvement at 48 h, or meningitis/severe sepsis	**Ceftriaxone**	**80–100 mg/kg** (max 4 g)	IV/IM	Once daily	Per diagnosis (7–10 days)
Specific infections	Malaria (artesunate/ACT), TB, dysentery, giardia (**metronidazole 7.5 mg/kg 8-hourly for 7 days**), meningitis	Per topic			
HIV-positive or exposed	**Co-trimoxazole** prophylaxis	Per HIV guidelines			

- **Measles–rubella vaccine** if ≥ 6 months and unvaccinated (or vaccinated before 9 months); delay if shocked.
- **Deworming** in the rehabilitation phase: **albendazole** 200 mg (12–23 months) or 400 mg (≥ 2 years) single dose (UCG: mebendazole 100 mg twice daily for 3 days, 7 days after admission, in endemic areas).
- **HIV**: start **ART as soon as possible after stabilisation** of the metabolic complications (UCG).
- If still anorexic after 7 days of antibiotics, continue for **10 days** total and reassess fully.

Step 6: Micronutrients

- **No iron in stabilisation.** Start **iron 3 mg/kg/day after 2 days on F-100**. **Never add iron to RUTF** (it already contains iron).

- F-75, F-100 and RUTF already contain **vitamin A, folic acid, zinc and copper**. If **not** on pre-mixed therapeutic foods: **folic acid 5 mg on day 1, then 1 mg daily**; **zinc 2 mg/kg/day**; **copper 0.3 mg/kg/day**; multivitamin syrup 5 mL daily, for at least 2 weeks.
- **Vitamin A only if eye signs or recent measles** (or not on therapeutic foods): **days 1, 2 and 14**: **< 6 months 50,000 IU; 6–12 months 100,000 IU; > 12 months 200,000 IU** (days 1 and 14 only if a dose was given at the referring centre).

Step 7: Cautious feeding (stabilisation) with F-75

Days	Frequency	Volume per feed	Volume per day
1–2	**2-hourly** (12 feeds)	**11 mL/kg**	**130 mL/kg**
3–5	**3-hourly** (8 feeds)	**16 mL/kg**	**130 mL/kg**
≥ 6	**4-hourly** (6 feeds)	**22 mL/kg**	**130 mL/kg**

- **Energy 100 kcal/kg/day (do not exceed)**; protein **1–1.5 g/kg/day**; fluid **130 mL/kg/day**, or **100 mL/kg/day if severe (+++) oedema**.
- Feed from a **cup and spoon** (syringe/dropper if weak); **never bottle**; **never leave the child alone** to feed (aspiration).
- **NG tube** if the child takes **≤ 80%** of the amount at **two consecutive feeds**.
- **Night feeds are essential.** Continue breastfeeding (breast first, then F-75).
- If vomiting: offer the amount vomited again; if persistent, **half the volume twice as often**.
- If **osmotic diarrhoea** with commercial F-75: use **cereal-based F-75**.
- Record feeds offered and left, vomits, stools and **daily weight**.

Step 8: Catch-up growth (transition, then rehabilitation)

Ready for transition: **appetite returned**, no hypoglycaemia (metabolically stable), oedema **reduced or gone**.

1. **Change gradually over 2–3 days.** Replace F-75 with the **same volume of F-100** (100 kcal and 2.9 g protein per 100 mL) for **2 days**, *or* start **RUTF** in small frequent meals (top up with F-75 if the child cannot finish; if less than half the RUTF is taken in 12 h, go back to F-75 for 1–2 days).
2. On **day 3**, increase each F-100 feed by **10 mL** until the child **leaves some** (about **200 mL/kg/day**).
3. Rehabilitation targets: **150–220 kcal/kg/day** and **4–6 g protein/kg/day**; frequent feeds, unlimited amounts. **F-100 is never sent home**: switch to RUTF.
4. **Watch for heart failure**: if **pulse rises by ≥ 25/min and RR by ≥ 5/min** on two successive **4-hourly** readings: **reduce to 100 mL/kg/day for 24 h**, then **115 mL/kg/day for 24 h**, then **130 mL/kg/day for 48 h**, then increase by 10 mL per feed again.
5. **Return to stabilisation** if appetite fails, oedema appears or increases, new medical problems, overload, or significant refeeding diarrhoea.
6. Monitor **weight gain** (worked example 2): aim for **> 10 g/kg/day**.

Step 9: Sensory stimulation and emotional support

- Tender loving care, a cheerful environment, **structured play 15–30 min/day**, physical activity when able, and as much **mother involvement** as possible (feeding, bathing, comforting). Support the **mother's** mental health (WHO 2023 A8, C2).

Step 10: Prepare for follow-up and discharge

- **Transfer to OTC** (WHO 2023 B3): **no danger signs for 24–48 h**, **medical problems resolved**, **oedema no longer +++ and resolving**, **good appetite** (passes the appetite test, eats > 80% of RUTF ration), not losing weight, caregiver linked to OTC. **Not** based on a target WHZ or MUAC.
- **Exit from the programme** (cured):

Guideline	Criteria
UCG 2023 / IMAM	**WHZ ≥ −2 OR MUAC ≥ 125 mm, and no oedema for at least 2 weeks**; use the **same indicator as at admission** (if admitted with oedema only, either may be used)
WHO 2023	**WHZ ≥ −2 AND MUAC ≥ 125 mm** on **2 consecutive visits**, and **no oedema for 2 consecutive visits**; % weight gain **not** used

- **Follow-up after discharge (UCG)**: at **1 week, 2 weeks, 1 month, 3 months and 6 months**, then twice a year until 3 years; monthly growth monitoring until 2 years; vaccines, vitamin A, deworming; play at home; **feed at least 5 times a day** with energy-dense food (100 kcal and 2–3 g protein per 100 g).

> UGANDA CONTEXT
>
> **Exit criteria and RUTF dose: Uganda vs WHO 2023.** Ugandan wards and the UCG 2023 still discharge on **one** indicator (the admission one) and dose RUTF at about **200 kcal/kg/day**. The **WHO 2023 wasting guideline** asks for **both** WHZ ≥ −2 **and** MUAC ≥ 125 mm on two visits and suggests **150–185 kcal/kg/day** (reducible to 100–130 once no longer severely wasted). Use the UCG/IMAM rules in Uganda, and mention WHO 2023 in an exam or viva.

E. Associated conditions

Condition	Treatment
Severe anaemia	Transfuse in the first 24 h **only if Hb < 4 g/dL, or 4–6 g/dL with respiratory distress**: **whole blood 10 mL/kg over 3 h** + **furosemide 1 mg/kg** at the start; **packed cells 10 mL/kg** if heart failure; monitor pulse, RR, liver every 15 min; slow down if RR rises by 5 or HR by 25; **stop and give furosemide** if crackles or enlarging liver
Corneal ulceration / keratomalacia	**Vitamin A** (days 1, 2, 14); **chloramphenicol or tetracycline eye drops** 4 times daily for 3–5 days; **atropine 1% drops** 3 times daily for 3–5 days; saline-soaked pads and bandage
Skin lesions (kwashiorkor dermatosis)	**0.01% potassium permanganate** soaks 10 min/day; **barrier cream** (zinc and castor oil or petroleum jelly); nystatin or gentian violet for sores; leave nappies off; zinc (in feeds)
Persistent diarrhoea	Giardia → metronidazole 7.5 mg/kg 8-hourly for 7 days; lactose intolerance (rare) → yoghurt or lactose-free formula; osmotic diarrhoea → cereal-based F-75
TB	Treat per national guideline (see TB topic); nutrition recovery will stall without it
HIV	ART after stabilisation; co-trimoxazole; slower recovery expected

F. Infants under 6 months (WHO 2023)

1. **Assess the mother–infant pair**: breastfeeding (attachment, positioning, suckling), perceived low supply, maternal health and **depression**.
2. **Admit** if danger signs, acute medical problems, **oedema** or **recent weight loss**.
3. **Breastfeeding is the treatment**: support the mother to breastfeed; if not breastfed, help her **relactate**; otherwise wet nursing.
4. **Supplementary suckling**: a fine NG tube taped beside the nipple delivers supplement while the baby suckles, stimulating supply. Supplement with **expressed breast milk**, or **infant formula, F-75 or diluted F-100** (no oedema), or **infant formula or F-75** (with oedema). **Never full-strength F-100** in unstable infants or with oedema, diarrhoea or dehydration (high renal solute load → hypernatraemia).
5. Transfer to outpatient care when no danger signs for **48 h**, problems resolved, oedema resolving, good appetite, and **satisfactory weight gain for 2–3 days**.

G. Refeeding syndrome

- **What**: dangerous falls in **phosphate, potassium and magnesium** (plus fluid retention, heart failure, hyperglycaemia) when feeding restarts after starvation; insulin drives these electrolytes into cells.
- **Signs**: weakness, breathing difficulty, oedema, **heart failure**, arrhythmia, confusion, seizures, sudden death in the first week.
- **Prevention**: start with **F-75** (low energy, low protein, contains phosphate, K and Mg), **do not exceed 100 kcal/kg/day** in stabilisation, increase gradually, give the electrolyte mix, monitor pulse, RR, weight and (where possible) phosphate.
- **Treatment**: reduce feeds (100 mL/kg/day), correct electrolytes, treat heart failure, then build up slowly.

H. Setting-specific plan

Level	Role
VHT / community	MUAC and oedema screening; refer red MUAC or oedema; counsel on feeding
HC II / HC III (OTC site)	Classify; appetite test; **OTC with RUTF and amoxicillin**; MAM counselling; growth monitoring; **pre-referral care** for complicated SAM: sugar water, warmth, first dose of ampicillin + gentamicin, ReSoMal if dehydrated, refer
HC IV / district hospital (ITC)	10 steps, F-75/F-100, ReSoMal, IV antibiotics, transfusion where available
RRH / NRH (Mulago Mwanamugimu, Mengo)	Complex cases: HIV/TB co-infection, electrolytes, blood cultures, ICU, specialist dietitians, infants under 6 months

Complications

Complication	Notes
Hypoglycaemia, hypothermia	Most common causes of early death; usually mean sepsis
Sepsis, pneumonia, UTI, TB, measles	Often without fever
Heart failure	From IV fluids, rapid ReSoMal, transfusion, or F-100 too early
Refeeding syndrome	Hypophosphataemia, hypokalaemia, hypomagnesaemia in week 1
Electrolyte disorders	Hypokalaemia, hypomagnesaemia, dilutional hyponatraemia
Severe anaemia	Iron, folate, malaria, HIV, worms
Xerophthalmia and blindness	Vitamin A deficiency, worse with measles
Skin breakdown and infection	Kwashiorkor dermatosis, candida
Persistent diarrhoea	Mucosal atrophy, lactose intolerance, giardia
Long term	**Stunting**, cognitive and developmental delay, relapse, later obesity and NCD risk, death after discharge

Ward-round & exam pearls

Model case presentation:

> "Aisha is a **14-month-old girl from Kawempe**, brought by her mother with **swelling of both feet for 2 weeks**, poor appetite and loose stools. She was **exclusively breastfed to 4 months**, then given thin maize porridge 2 times a day; she stopped breastfeeding at 11 months when her mother became pregnant again. She is **HIV-exposed**: mother on TLD, Aisha had a negative DNA PCR at 6 weeks but not at 9 months. **Immunisation**: up to date to 14 weeks; **missed measles–rubella at 9 months**. Development: sitting, not yet crawling. On examination: **miserable and apathetic**, **oedema ++** (feet and lower legs), flaky-paint dermatosis on the thighs, sparse pale hair. **Weight 6.4 kg, length 70 cm (WLZ −2.4, LAZ −2.7, WAZ −3.3)**, **MUAC 11.8 cm**. Axillary temp **35.8 °C**, RBS **3.4 mmol/L**, HR 128, RR 36, no chest signs, liver 3 cm, no gallop. **Hb 7.2 g/dL**, malaria RDT negative, **urine protein negative**. **Appetite test failed** (ate 2 teaspoons of RUTF in 30 min). **Summary**: **complicated SAM with nutritional oedema (++) (kwashiorkor)** and poor appetite, moderate stunting, missed MR vaccine, HIV-exposed with unknown status. **Plan**: admit to the nutrition unit, keep warm, **F-75 130 mL/kg/day 2-hourly (70 mL per feed)** including nights, **ampicillin + gentamicin**, ReSoMal only if watery stools, skin care, **DNA PCR**, **MR vaccine**, **no iron, no diuretics**, daily weight and oedema check; counsel on complementary feeding and family planning."

Likely viva questions (with answers)

1. *Define SAM in a 2-year-old.* WHZ < −3, or MUAC < 115 mm, or bilateral pitting oedema.
2. *Why not give furosemide for kwashiorkor oedema?* Oedema reflects sodium retention, potassium depletion and capillary leak, not volume excess; diuretics cause hypovolaemia and electrolyte collapse. Oedema resolves with feeding and potassium.
3. *Why not give iron on admission?* Free iron promotes bacterial growth and oxidative damage while transferrin is low; start iron after 2 days on F-100 (never with RUTF).
4. *Why F-75 first?* Low protein, low sodium, low osmolality and enough energy to stop catabolism without overwhelming the slowed metabolism; prevents refeeding syndrome and heart failure.
5. *What is the appetite test and why does it matter?* RUTF offered for 30 min; eating at least ⅓ of a 92 g sachet = pass. Failure means complicated SAM and admission.
6. *Name the 10 steps.* Hypoglycaemia, hypothermia, dehydration, electrolytes, infection, micronutrients, cautious feeding, catch-up growth, sensory stimulation, preparing for follow-up.
7. *What does WHO 2023 change?* "Wasting and nutritional oedema" terminology; infants < 6 months "at risk of poor growth and development"; ReSoMal or low-osmolarity ORS for dehydration; RUTF 150–185 kcal/kg/day; exit on both WHZ and MUAC; transfer to outpatient care on clinical, not anthropometric, criteria; LNS for high-risk MAM.

OSCE tips: in "measure MUAC", explain to the mother, use the **left arm**, find the **midpoint** between acromion and olecranon with the arm bent at 90°, then let the arm hang, wrap the tape **snugly without pinching**, read to the **nearest mm**, and **state the colour and classification**. In "check for oedema", press **both feet** for **3 seconds** and look for pits. In "plot a growth chart", choose the **right sex and chart**, plot accurately, join the dots, and **interpret the trend**.

Mnemonics & memory aids

- **"115 / 125"**: MUAC red below 115, yellow to 125, green above.
- **"Minus 2 moderate, minus 3 severe."**
- **10 steps, "Hypo-Hypo-Dehydrated-Electrified-Infected Micro Cautious Catch-up Stimulating Follow-up"** or **"HHDEI-MCCSF"**: the first five are emergencies.
- **F-75 "11, 16, 22"** mL/kg per feed at 2-, 3- and 4-hourly feeds (all 130 mL/kg/day).
- **"No IV, no iron, no diuretics, no bottles, no skipping nights."**
- **Weight gain "5 and 10"**: < 5 poor, 5–10 moderate, > 10 good (g/kg/day).

- **RUTF: "92 grams, 500 kcal".**

High-yield summary

Numbers & doses to know

Item	Value
SAM (6–59 mo)	**WHZ < –3, MUAC < 115 mm, or bilateral pitting oedema**
MAM	WHZ –3 to < –2, MUAC **115–< 125 mm**
Infant < 6 mo at risk (WHO 2023)	WAZ < –2, WLZ < –3, **MUAC < 110 mm** (6 wk–6 mo), weight faltering, ineffective breastfeeding
Stunting / underweight	HAZ < –2 / WAZ < –2 (severe < –3)
Overweight / obesity (< 5 y)	WHZ > +2 / > +3
Appetite test pass	≥ **⅓ of a 92 g RUTF sachet** in 30 min
Hypoglycaemia	**< 3 mmol/L**: 50 mL 10% glucose/sugar water; IV **5 mL/kg 10% glucose**
Hypothermia	Axillary **< 35 °C**, rectal **< 35.5 °C**
ReSoMal	**5 mL/kg every 30 min × 2 h**, then **5–10 mL/kg/h** for 4–10 h
SAM shock	**15 mL/kg over 1 h**; blood **10 mL/kg over 3 h** if not improving
Extra K / Mg	**3–4 / 0.4–0.6 mmol/kg/day**
F-75	75 kcal, 0.9 g protein /100 mL; **130 mL/kg/day** (100 if oedema +++); **100 kcal/kg/day** max
F-75 schedule	**11 mL/kg 2-hourly** (d1–2), **16 mL/kg 3-hourly** (d3–5), **22 mL/kg 4-hourly** (d6+)
F-100	100 kcal, 2.9 g protein /100 mL; same volume × 2 days, then **+10 mL/feed**; up to ~**200 mL/kg/day**
Rehabilitation intake	**150–220 kcal/kg/day**, **4–6 g protein/kg/day**
RUTF	92 g = **500 kcal**; UCG ≈ 200 kcal/kg/day; WHO 2023 **150–185 kcal/kg/day**
Antibiotics, uncomplicated	**Amoxicillin 40 mg/kg 12-hourly × 5 days**
Antibiotics, complicated	**Ampicillin 50 mg/kg 6-hourly × 2 days → amoxicillin 25–40 mg/kg 8-hourly × 5 days + gentamicin 7.5 mg/kg daily × 7 days**
Iron	**3 mg/kg/day** after 2 days on F-100; **none with RUTF**
Folic acid (if not on therapeutic food)	**5 mg day 1, then 1 mg daily**
Vitamin A (eye signs/measles)	**50,000 / 100,000 / 200,000 IU** (< 6 mo / 6–12 mo / > 12 mo) days 1, 2, 14
Transfusion in SAM	Hb **< 4**, or **4–6 with distress**: whole blood **10 mL/kg over 3 h** + furosemide 1 mg/kg
Heart failure on feeds	HR **+25** and RR **+5** on 2 successive 4-hourly readings → feeds **100 mL/kg/day**
Weight gain	Good **> 10**, moderate **5–10**, poor **< 5 g/kg/day**
Exit (UCG)	WHZ ≥ –2 **or** MUAC ≥ 125 mm, no oedema × 2 weeks
Uganda (UDHS 2022)	Stunting **26%**, wasting **2.9%**, underweight **10.2%**

Classic exam traps

- **"Furosemide for kwashiorkor oedema."** Wrong: never.
- **"IV Plan C for a dehydrated child with SAM."** Wrong: ReSoMal orally/NG; IV only for shock (15 mL/kg/h).
- **"Iron syrup on admission for anaemia."** Wrong: no iron in stabilisation; none with RUTF.
- **"High-dose vitamin A to every child with SAM."** Wrong (since WHO 2013): only with eye signs, measles, or if not on therapeutic foods.
- **"No fever, so no antibiotics."** Wrong: all children with SAM get antibiotics.
- **"Weight-for-age < –3, so SAM."** Wrong: WAZ is underweight; SAM uses WHZ, MUAC or oedema.
- **"Discharge when weight has increased by 15%."** Wrong: % weight gain is not an exit criterion.
- **"Oedema in a malnourished child = kwashiorkor."** Check urine protein (nephrotic) and the heart and liver first.

Questions seniors ask

The child is hypothermic at 3 a.m. What do you do? Feed immediately (F-75), check glucose (treat if < 3), skin-to-skin with the mother, cover the head, warm the room, make sure antibiotics were given (consider escalating to ceftriaxone), and recheck the temperature in 30–60 minutes.

Why does this child need gentamicin when he has no fever? SAM suppresses fever and inflammatory signs; bacteraemia (often Gram-negative) is common and causes hypoglycaemia, hypothermia and death, so all complicated SAM gets broad-spectrum cover.

When is the child ready to move from F-75 to F-100 or RUTF? When appetite returns, there has been no hypoglycaemia (metabolically stable), and oedema is reducing or gone, usually within 2–7 days.

She is on F-100 and her pulse has risen from 120 to 150 and RR from 40 to 46 on two readings. What is happening? Early heart failure from too rapid an increase in feeds: reduce to 100 mL/kg/day for 24 h, then 115 for 24 h, then 130 for 48 h, then increase by 10 mL per feed; check for overload and anaemia.

He is 2 weeks into rehabilitation but gaining 3 g/kg/day. What will you look for? Feeding problems (feeds not given at night, diluted, eaten by siblings), hidden infection (TB, HIV, UTI, otitis, pneumonia), malabsorption or giardia, missed vitamin/mineral mix, and caregiver problems (depression, workload).

What is the difference between wasting and stunting, and which kills? Wasting (low WHZ, MUAC) is acute and carries the highest short-term mortality; stunting (low HAZ) is chronic, largely irreversible after 2 years, and limits development and adult health. Both often co-exist.

When should this child with HIV start ART? As soon as possible after the metabolic complications are stabilised (UCG 2023), usually within the first 1–2 weeks, with co-trimoxazole prophylaxis and TB screening.

SELF-TEST

Flashcards

29 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Criteria for SAM in children 6–59 months	**WHZ < –3** OR **MUAC < 115 mm** OR **bilateral pitting oedema** (any one).
2	Criteria for MAM	**WHZ –3 to < –2** OR **MUAC 115 to < 125 mm**, **no oedema**.
3	What do WHZ, HAZ and WAZ measure?	**WHZ** = wasting (acute); **HAZ/LAZ** = stunting (chronic); **WAZ** = underweight (mixed). Moderate < –2, severe < –3.
4	Grades of nutritional oedema	**+** both feet; **++** feet plus lower legs, hands or lower arms; **+++** generalised including the face.
5	Complicated SAM: admission criteria	SAM plus any **IMCI danger sign** or **medical complication**, **oedema +++**, **failed appetite test**, or age **< 6 months** (WHO 2023: infant at risk with danger signs, oedema or recent weight loss).
6	How is the appetite test done and passed (UCG)?	Offer RUTF in a quiet corner with clean water for **30 minutes**. **Pass** = eats at least **⅓ of a 92 g sachet** (or 3 teaspoons from a pot).
7	The WHO 10 steps for SAM	1 Hypoglycaemia 2 Hypothermia 3 Dehydration 4 Electrolytes 5 Infection 6 Micronutrients 7 Cautious feeding 8 Catch-up growth 9 Sensory stimulation 10 Prepare for follow-up.
8	Treatment of hypoglycaemia in SAM (conscious child)	**50 mL of 10% glucose or sugar water** orally/NG, then **F-75 every 30 min for 2 h**, then 2-hourly; antibiotics; recheck glucose at 30 min. Unconscious: **10% glucose 5 mL/kg IV**.
9	Hypothermia thresholds and treatment in SAM	Axillary **< 35 °C** / rectal **< 35.5 °C**. Feed, **kangaroo care**, cover head, blankets, warm room (no hot water bottles), antibiotics, check glucose.
10	Rehydration in SAM without shock	**ReSoMal 5 mL/kg every 30 min for 2 h**, then **5–10 mL/kg/h for 4–10 h** (alternate with F-75). No IV.
11	Shock in SAM	**15 mL/kg over 1 h** of RL with 5% glucose (or half-strength Darrow's / 0.45% saline with 5% glucose); repeat once if improving; if not, **whole blood 10 mL/kg over 3 h**.
12	Antibiotics in complicated SAM (UCG 2023)	**Ampicillin 50 mg/kg IV/IM 6-hourly × 2 days**, then **amoxicillin 25–40 mg/kg 8-hourly × 5 days**, **PLUS gentamicin 7.5 mg/kg once daily × 7 days**.
13	Antibiotic in uncomplicated SAM (UCG 2023)	**Amoxicillin 40 mg/kg orally twice daily for 5 days.**
14	F-75 composition and daily volume	**75 kcal and 0.9 g protein per 100 mL. 130 mL/kg/day** (100 mL/kg/day if severe oedema); max 100 kcal/kg/day.
15	F-75 feeding schedule	Days 1–2: **2-hourly, 11 mL/kg/feed**; days 3–5: **3-hourly, 16 mL/kg**; day 6+: **4-hourly, 22 mL/kg**.
16	When to use an NG tube for F-75	If the child takes **≤ 80%** of the feed at **two consecutive feeds**.
17	Signs that a child is ready for transition to F-100/RUTF	**Return of appetite**, **no hypoglycaemia** (metabolically stable), **oedema reduced or gone**.
18	How do you transition from F-75 to F-100?	Same volume of **F-100 for 2 days**, then **increase each feed by 10 mL** until some is left (about 200 mL/kg/day). Never switch all at once; F-100 never goes home.
19	Signs of heart failure during transition and what to do	Pulse **+25/min** and RR **+5/min** on 2 successive 4-hourly readings → **100 mL/kg/day × 24 h**, then 115 × 24 h, then 130 × 48 h, then +10 mL/feed.

#	Question	Answer
20	Weight gain categories in rehabilitation	**Good > 10 g/kg/day**; moderate **5–10**; **poor < 5** (full reassessment).
21	Iron in SAM	**None in stabilisation.** Start **3 mg/kg/day after 2 days on F-100**. **Never with RUTF** (already contains iron).
22	When is high-dose vitamin A given in SAM?	Only with **eye signs of deficiency** or **recent measles** (or if not on therapeutic foods): days 1, 2 and 14 — **50,000 IU (< 6 mo), 100,000 IU (6–12 mo), 200,000 IU (> 12 mo)**.
23	RUTF energy content and WHO 2023 dose	One **92 g sachet = 500 kcal**. WHO 2023: **150–185 kcal/kg/day** (may reduce to 100–130 once no longer severely wasted). UCG table ≈ 200 kcal/kg/day.
24	Exit criteria: UCG 2023 vs WHO 2023	**UCG**: WHZ ≥ −2 **or** MUAC ≥ 125 mm (same indicator as admission) and no oedema for 2 weeks. **WHO 2023**: WHZ ≥ −2 **and** MUAC ≥ 125 mm on 2 consecutive visits, no oedema × 2 visits.
25	Uganda child nutrition (UDHS 2022)	**Stunting 26%**, **wasting 2.9%**, **underweight 10.2%**; worst in **Karamoja**.
26	Marasmus vs kwashiorkor	**Marasmus**: severe wasting, old man's face, baggy pants, alert and hungry, no oedema. **Kwashiorkor**: bilateral pitting oedema, apathy, flaky-paint dermatosis, sparse pale hair, fatty liver, poor appetite.
27	Transfusion criteria in SAM	Only if **Hb < 4 g/dL**, or **4–6 g/dL with respiratory distress**: **whole blood 10 mL/kg over 3 h** + **furosemide 1 mg/kg**; packed cells if heart failure.
28	WHO 2023 criteria for an infant < 6 months 'at risk of poor growth and development'	E.g. **WAZ < −2**, **WLZ < −3**, **MUAC < 110 mm** (6 weeks–6 months), **failure to gain weight** on 2 measurements, **ineffective breastfeeding**; admit if danger signs, oedema or recent weight loss.
29	Normal growth benchmarks in infancy	Birth weight **doubles at 4–5 months**, **triples at 12 months**; length 50 → **75 cm** at 1 year; head circumference 35 → **47 cm** at 1 year.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy (8.2 kg, height 82 cm) is seen at a HC III. MUAC 11.2 cm, no oedema. He is alert, has no danger signs, no cough or diarrhoea, and eats a whole third of a RUTF sachet in 15 minutes. What is the MOST appropriate management?

A Admit for F-75 stabilisation

B Counsel on feeding and review in 30 days

C Outpatient therapeutic care: RUTF about 3 sachets/day, amoxicillin 40 mg/kg twice daily for 5 days, weekly review

D Give ferrous sulphate, vitamin A 200,000 IU and RUTF

E Refer to the regional referral hospital for investigations

2. A 16-month-old girl with SAM (MUAC 10.6 cm) is admitted at 22:00. She is lethargic but rousable, axillary temperature 34.8 °C, RBS 2.2 mmol/L. She can swallow. Which is the MOST appropriate first action?

A 50 mL of 10% glucose (or sugar water) orally/NG, start F-75, keep her warm skin-to-skin, and give IV antibiotics

B Place her under a radiant heater and wait until morning to feed

C 10% glucose 5 mL/kg IV and 20 mL/kg Ringer's lactate bolus

D Give F-100 to provide more energy

E Hot water bottles around her and oral iron

3. An 18-month-old with oedema ++ and MUAC 12.0 cm is admitted. The intern prescribes furosemide 1 mg/kg to 'clear the oedema before feeding'. What is the BEST response?

A Agree, and add albumin infusion

B Agree, but reduce the dose to 0.5 mg/kg

C Give spironolactone instead

D Stop the diuretic: nutritional oedema is not treated with diuretics; it resolves with F-75, potassium and magnesium

E Restrict feeds to 50 mL/kg/day until oedema resolves

4. A 3-year-old (10 kg) with SAM and watery diarrhoea has sunken eyes, slow skin pinch, warm hands, CRT 2 s and a good radial pulse. Which rehydration plan is MOST appropriate?

A Ringer's lactate 100 mL/kg over 3 hours

B Standard ORS 750 mL over 4 hours

C Ringer's lactate with 5% glucose 150 mL over 1 hour

D No fluids, because the eye and skin signs are due to wasting

E ReSoMal 50 mL every 30 minutes for 2 hours, then 50–100 mL/h for up to 10 hours, monitoring pulse, RR and liver

5. A 13-month-old (6.0 kg) with complicated SAM has just been admitted. No oedema. Which F-75 prescription is correct for days 1–2?

A 130 mL every 4 hours

B 66 mL every 2 hours, day and night

C 100 mL every 3 hours

D 150 mL every 3 hours

E 40 mL every 2 hours during the day only

6. A 2-year-old with SAM is on day 4. He finishes his F-75 eagerly, oedema has gone from ++ to +, and there has been no hypoglycaemia since day 1. What is the MOST appropriate next step?

A Continue F-75 for another week
B Change all feeds to F-100 at 200 mL/kg/day today
C Discharge home on F-100
D Start transition: same volume of F-100 (or RUTF in small frequent meals) for 2 days, then increase each feed by 10 mL
E Start iron and vitamin A

7. Two days after starting F-100, a 20-month-old's pulse rises from 118 to 146/min and RR from 38 to 45/min, sustained on two 4-hourly readings. The liver edge has moved 1 cm lower. What is the MOST appropriate action?

Advanced

A Reduce feeds to 100 mL/kg/day for 24 h, then 115 for 24 h, then 130 for 48 h, and reassess
B Give a 10 mL/kg saline bolus
C Switch to RUTF and discharge
D Increase feeds to provide more energy
E Give oral iron because of anaemia

8. A 2-year-old with SAM has gained weight from 7.0 kg to 7.3 kg over 3 days on F-100. What is his rate of weight gain and how should it be interpreted?

A About 3 g/kg/day: poor, needs full reassessment
B About 7 g/kg/day: moderate, check intake and look for infection
C About 14 g/kg/day: good
D About 21 g/kg/day: excessive, reduce feeds
E About 100 g/day: cannot be interpreted without length

9. A 15-month-old with complicated SAM (MUAC 10.4 cm, no oedema) is on F-75. Hb 3.6 g/dL. She has no respiratory distress and no signs of heart failure. What is the MOST appropriate management of the anaemia?

Advanced

A Oral iron 3 mg/kg/day now
B Whole blood 20 mL/kg over 1 hour
C No transfusion; give folic acid and wait for rehabilitation
D Packed cells 20 mL/kg over 4 hours
E Whole blood 10 mL/kg over 3 hours with furosemide 1 mg/kg at the start, monitoring every 15 minutes

10. An 11-month-old girl's weight-for-age line on her child health card ran along the median until 6 months, then flattened over 4 months and now lies just below −2. MUAC 12.2 cm, no oedema, WLZ −2.3. She is well with a good appetite. Which is the MOST appropriate response?

A Reassure: she is still breastfeeding
B Classify as MAM, take a feeding history and counsel on complementary feeding, check for illness and HIV exposure, and review in 2–4 weeks
C Admit for F-75
D Start RUTF and amoxicillin
E Advise stopping breastfeeding so she eats more porridge

11. A 3-year-old boy has bilateral pitting oedema up to the knees and puffy eyelids. He looks well nourished (MUAC 14.6 cm), is playful, with no skin or hair changes. Urine dipstick: protein 4+, no blood. What is the MOST likely diagnosis?

A Kwashiorkor
B Nephrotic syndrome
C Acute post-streptococcal glomerulonephritis
D Congestive cardiac failure
E Marasmic kwashiorkor

12. A 4-month-old girl (4.0 kg, birth weight 3.1 kg) is brought because she 'is not growing'. She is breastfed but the mother has sore nipples and feeds her water and sugar solution 'because the milk is little'. No danger signs, no oedema, WLZ −2.8. Mother is tearful. According to WHO 2023, what is the MOST appropriate approach? Advanced

A Admit and give full-strength F-100
B Start RUTF one sachet daily
C Assess the mother–infant pair, correct breastfeeding attachment and stop water/sugar, screen the mother for depression, and follow up closely as an outpatient
D Switch to formula feeding exclusively
E Give amoxicillin and iron syrup

13. A 2-year-old with SAM was admitted with MUAC 10.9 cm. After 7 weeks of OTC his MUAC is 12.8 cm on two consecutive visits, no oedema for 3 weeks, WHZ −2.3. Which statement is MOST accurate? Advanced

A He meets UCG 2023 exit criteria (MUAC ≥ 125 mm, the admission indicator, no oedema for 2 weeks) but not WHO 2023 criteria, which also require WHZ ≥ −2
B He cannot be discharged until he has gained 15% of his admission weight
C He meets both UCG and WHO 2023 exit criteria
D He should be readmitted for F-100
E He should be discharged only when WAZ is above −2

14. On day 3 of F-75, a 14-month-old with SAM becomes weak and breathless, with new oedema and a pulse of 165. Serum phosphate is 0.4 mmol/L, potassium 2.8 mmol/L. The nurse reports she had been giving 'extra feeds because the child was hungry'. What is the MOST likely diagnosis? Advanced

A Hospital-acquired pneumonia
B Hypoglycaemia
C Refeeding syndrome
D Kwashiorkor relapse
E Malaria

15. A 10-month-old with SAM has Bitot's spots and a hazy right cornea. She is on F-75. Which management of the eyes is MOST appropriate?

A Vitamin A 100,000 IU on days 1, 2 and 14, chloramphenicol eye drops 4 times daily and atropine 1% drops, and pad the eye
B No vitamin A because F-75 contains it
C Vitamin A 200,000 IU single dose only
D Topical steroid drops for the corneal haze
E Refer to ophthalmology after nutritional rehabilitation

Answers & explanations

1. C · Outpatient therapeutic care: RUTF about 3 sachets/day, amoxicillin 40 mg/kg twice daily for 5 days, weekly review

C is correct: MUAC < 115 mm = SAM; he passes the appetite test and has no complications, so he is treated as an outpatient with RUTF (UCG table: 7–8.4 kg = 3 sachets/day) and amoxicillin 40 mg/kg twice daily for 5 days, with 1–2-weekly follow-up. A is wrong: inpatient care is for complicated SAM or a failed appetite test. B is wrong: that is the plan for MAM; SAM needs therapeutic food and antibiotics. D is wrong: RUTF already contains iron and vitamin A; extra iron is not given, and high-dose vitamin A is only for eye signs or measles. E is wrong: uncomplicated SAM is managed at HC level.

2. A · 50 mL of 10% glucose (or sugar water) orally/NG, start F-75, keep her warm skin-to-skin, and give IV antibiotics

A is correct: steps 1 and 2 together: treat hypoglycaemia with 50 mL of 10% glucose or sugar water orally/NG (she can swallow), start F-75 (every 30 min for 2 h), warm her by kangaroo care, and give antibiotics because hypoglycaemia and hypothermia usually mean sepsis. B is wrong: delaying feeds causes recurrent hypoglycaemia and death; night feeds are essential. C is wrong: IV glucose is for unconscious or convulsing children, and a 20 mL/kg bolus is dangerous in SAM without shock. D is wrong: F-100 in stabilisation risks refeeding syndrome and heart failure. E is wrong: hot water bottles cause burns, and iron is never given in stabilisation.

3. D · Stop the diuretic: nutritional oedema is not treated with diuretics; it resolves with F-75, potassium and magnesium

D is correct: oedema in SAM reflects sodium retention, potassium depletion and capillary leak; diuretics cause hypovolaemia and electrolyte collapse and can kill. It resolves with cautious feeding (F-75), extra K and Mg and treating infection. A is wrong: albumin and diuretics are not used. B is wrong: any diuretic dose is inappropriate here. C is wrong: spironolactone is also a diuretic with no role. E is wrong: F-75 is given at 130 mL/kg/day (100 mL/kg/day only if oedema +++); starving the child worsens outcome.

4. E · ReSoMal 50 mL every 30 minutes for 2 hours, then 50–100 mL/h for up to 10 hours, monitoring pulse, RR and liver

E is correct: SAM with dehydration but not shocked = ReSoMal 5 mL/kg every 30 min for 2 h, then 5–10 mL/kg/h for 4–10 h, with close monitoring for overload. A is wrong: Plan C IV fluids are not used in SAM without shock. B is wrong: standard Plan B volumes and sodium are unsafe (low-osmolarity ORS is only the fallback, given slowly). C is wrong: 15 mL/kg IV over 1 h is the shock protocol; he is not shocked. D is wrong: although the signs are unreliable in SAM, a history of watery diarrhoea means he should be assumed to have some dehydration and given ReSoMal.

5. B · 66 mL every 2 hours, day and night

B is correct: 130 mL/kg/day = 780 mL/day; days 1–2 are 2-hourly (12 feeds) at 11 mL/kg = 66 mL per feed, including night feeds. A is wrong: 22 mL/kg 4-hourly is the day 6+ schedule, and 130 mL is too large per feed. C is wrong: 16 mL/kg 3-hourly is for days 3–5 (≈ 96 mL). D is wrong: 150 mL 3-hourly = 1200 mL/day = 200 mL/kg/day, rehabilitation volumes that risk heart failure. E is wrong: the volume is too low and omitting night feeds causes hypoglycaemia.

6. D · Start transition: same volume of F-100 (or RUTF in small frequent meals) for 2 days, then increase each feed by 10 mL

D is correct: appetite has returned, he is metabolically stable and oedema is reducing, so he is ready for a gradual 2–3-day transition (same volume of F-100 for 2 days, then +10 mL per feed; or RUTF). A is wrong: prolonging F-75 delays catch-up growth. B is wrong: switching abruptly to full rehabilitation volumes risks heart failure and refeeding syndrome. C is wrong: F-100 is never given to take home (RUTF is used for outpatient rehabilitation). E is wrong: iron starts after 2 days on F-100 (never with RUTF), and vitamin A is only for eye signs or measles.

7. A · Reduce feeds to 100 mL/kg/day for 24 h, then 115 for 24 h, then 130 for 48 h, and reassess

A is correct: a sustained rise of ≥ 25 in pulse and ≥ 5 in RR with an enlarging liver signals early heart failure from feed volume (UCG/WHO); reduce to 100 mL/kg/day and build up slowly, and look for other causes (anaemia, infection, refeeding). B is wrong: fluid worsens heart failure. C is wrong: an unstable child must not be discharged. D is wrong: more volume worsens the failure. E is wrong: iron does not treat heart failure.

8. C · About 14 g/kg/day: good

C is correct: gain 300 g ÷ 3 days = 100 g/day; average weight 7.15 kg; 100 ÷ 7.15 ≈ 14 g/kg/day = good (> 10). A is wrong: miscalculation; < 5 g/kg/day would be poor. B is wrong: 5–10 is moderate; his rate is higher. D is wrong: 21 g/kg/day is not the result and is not by itself a reason to cut feeds without signs of overload. E is wrong: g/kg/day is calculated from weights alone.

9. E · Whole blood 10 mL/kg over 3 hours with furosemide 1 mg/kg at the start, monitoring every 15 minutes

E is correct: in SAM, transfuse in the first 24 h only if Hb < 4 g/dL (or 4–6 with respiratory distress), giving smaller volumes more slowly: whole blood 10 mL/kg over 3 h with furosemide 1 mg/kg and close monitoring. A is wrong: iron is not given in stabilisation. B is wrong: 20 mL/kg over 1 h risks heart failure in SAM. C is wrong: Hb < 4 meets the transfusion threshold. D is wrong: the volume is too large for a child with SAM.

10. B · Classify as MAM, take a feeding history and counsel on complementary feeding, check for illness and HIV exposure, and review in 2–4 weeks

B is correct: WLZ −2.3 and MUAC 12.2 cm = MAM; growth faltering began with complementary feeding, so assess the diet and illnesses (malaria, diarrhoea, worms, HIV), counsel and review soon (consider LNS if high-risk per WHO 2023). A is wrong: a flattening line is the early warning growth monitoring is designed to catch. C is wrong: she does not have SAM or complications. D is wrong: RUTF and antibiotics are for SAM. E is wrong: breastfeeding should continue to 2 years and beyond.

11. B · Nephrotic syndrome

B is correct: heavy proteinuria with periorbital and leg oedema in a well-nourished, playful child points to nephrotic syndrome; always dip the urine before labelling oedema as nutritional. A is wrong: kwashiorkor has negative or trace urine protein, apathy and skin/hair changes. C is wrong: AGN causes haematuria, hypertension and oliguria. D is wrong: there are no cardiac signs described. E is wrong: there is no wasting.

12. C · Assess the mother–infant pair, correct breastfeeding attachment and stop water/sugar, screen the mother for depression, and follow up closely as an outpatient

C is correct: WHO 2023 treats this as an infant < 6 months at risk of poor growth and development without admission criteria; management centres on the mother–infant pair: breastfeeding support (attachment, frequency), stopping other fluids, maternal mental health, and close outpatient follow-up. A is wrong: full-strength F-100 is never given to young infants (renal solute load), and she has no admission criteria. B is wrong: RUTF is not for infants under 6 months. D is wrong: breastfeeding is the treatment; replacement feeding is only if there is no realistic prospect of breastfeeding. E is wrong: there is no indication for these.

13. A · He meets UCG 2023 exit criteria (MUAC ≥ 125 mm, the admission indicator, no oedema for 2 weeks) but not WHO 2023 criteria, which also require WHZ ≥ −2

A is correct: UCG 2023/IMAM uses the admission indicator (MUAC ≥ 125 mm) with no oedema for 2 weeks; WHO 2023 requires both WHZ ≥ −2 and MUAC ≥ 125 mm on 2 consecutive visits, and his WHZ is −2.3. B is wrong: percentage weight gain is no longer an exit criterion. C is wrong: he fails the WHO WHZ criterion. D is wrong: he is improving and well; readmission is not indicated. E is wrong: WAZ is not an exit criterion.

14. C · Refeeding syndrome

C is correct: rapid increase in feeding after starvation drives phosphate, potassium and magnesium into cells (insulin surge), causing weakness, fluid retention and heart failure in the first week. Reduce feeds to 100 mL/kg/day, correct electrolytes and build up slowly. A is wrong: pneumonia does not explain the profound hypophosphataemia after excess feeding. B is wrong: glucose is not reported low and hypoglycaemia does not cause hypophosphataemia. D is wrong: oedema here is part of refeeding, not a dietary relapse. E is wrong: malaria does not explain the electrolyte pattern (but always test).

15. A · Vitamin A 100,000 IU on days 1, 2 and 14, chloramphenicol eye drops 4 times daily and atropine 1% drops, and pad the eye

A is correct: eye signs of vitamin A deficiency need high-dose vitamin A on days 1, 2 and 14 (100,000 IU at 6–12 months) plus corneal care: antibiotic drops, atropine and a protective pad. Corneal clouding is an emergency (keratomalacia). B is wrong: therapeutic feeds contain maintenance vitamin A, not treatment doses for xerophthalmia. C is wrong: 200,000 IU is the dose for > 12 months and a single dose is insufficient. D is wrong: steroids risk corneal perforation and infection. E is wrong: delay risks blindness; treat now and involve eye services early.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Kwashiorkor at Mulago's nutrition unit

Peter, a 22-month-old boy from Kalangala, is referred to Mwanamugimu Nutrition Unit with 3 weeks of swelling of the feet and face, peeling skin and poor appetite. His mother had a new baby 3 months ago and Peter was weaned abruptly onto cassava porridge.

1. What will you measure and look for first?

> **ETAT danger signs**, then **weight (8.1 kg), length (80 cm), MUAC (12.4 cm)**, **oedema** (feet, legs, hands and **face = +++**), axillary **temperature (35.6 °C)**, **RBS (3.8 mmol/L)**, hydration, circulation (CRT 2 s), eyes (no xerosis), **skin** (flaky-paint dermatosis, raw areas on thighs), hair (sparse, pale), chest, **liver (4 cm, soft)**, signs of heart failure, **urine dipstick (protein trace)**.

2. Classify him and explain why he needs admission.

> **Complicated SAM with nutritional oedema +++ (kwashiorkor)**. Oedema +++ alone is an admission criterion (WHO 2023 B2), and he has poor appetite and skin breakdown. His MUAC (12.4 cm) is only MAM-range: **oedema overrides anthropometry** (weight is inflated by fluid).

3. Write the first 24-hour orders.

> - Keep warm (kangaroo with mother, hat, blankets); temp 2-hourly
> - **F-75 at 100 mL/kg/day** (oedema +++) = 810 mL/day: **≈ 68 mL 2-hourly**, day and night; NG if < 80% taken twice
> - **Ampicillin 400 mg (50 mg/kg) IV 6-hourly × 2 days + gentamicin 60 mg (7.5 mg/kg) IV once daily × 7 days**, then amoxicillin
> - **No diuretics, no iron, no IV fluids**; low salt
> - Electrolyte/mineral mix in feeds (K 3–4, Mg 0.4–0.6 mmol/kg/day)
> - Skin: **0.01% potassium permanganate** soaks, barrier cream, nystatin for candida
> - RBS 4-hourly initially; daily weight and oedema grade
> - **HIV test**, malaria RDT, Hb, urine culture; measles–rubella vaccine check

4. On day 5 his weight has fallen from 8.1 to 7.4 kg and oedema is +. His mother is worried. What do you tell her?

> Weight loss here is **good news**: he is **losing oedema fluid** as his cells recover potassium and sodium is excreted. Real weight gain will begin once oedema has gone and he moves to F-100 or RUTF. Praise her for feeding him through the night.

5. On day 7 he is hungry, oedema has gone, and he passes the appetite test. What next?

> **Transition** over 2–3 days to **RUTF** (or F-100 same volume × 2 days then +10 mL/feed), monitor pulse and RR 4-hourly for heart failure. Start **sensory stimulation** (play sessions). Once eating > 80% of RUTF, no danger signs for 24–48 h and medical problems resolved, **transfer to OTC** near home with a 2-week RUTF supply.

6. How will you prevent relapse?

> Counsel on **complementary feeding** (thick porridge enriched with oil, groundnut paste, fish/silverfish (mukene), beans, vegetables; 3–4 meals + 1–2 snacks; own plate), hygiene, **family planning** (short birth interval caused abrupt weaning), link to **OTC follow-up** and community support, **vitamin A and deworming** schedule, growth monitoring monthly, and screen the mother for depression and food insecurity.

Take-away: Oedema +++ means admission even when MUAC looks reassuring; feed F-75 at 100 mL/kg/day, give antibiotics, never diuretics, expect weight to fall as oedema clears, and tackle the social cause.

Case 2: Severe wasting with HIV and pneumonia

Esther, a 9-month-old girl, is brought to a regional referral hospital with cough for 3 weeks, fast breathing and loose stools. Her mother tested HIV-positive in this pregnancy but stopped ART after delivery. Esther weighs 5.2 kg, length 68 cm.

1. What is your rapid assessment?

> RR 62, **chest indrawing**, SpO_2 88% → **oxygen**. Temp 35.4 °C (axillary), **RBS 2.6 mmol/L**. Hands warm, CRT 2 s. **MUAC 10.2 cm, WLZ < −3**, no oedema. Oral thrush, generalised lymphadenopathy. Classification: **complicated SAM (severe wasting)** with **severe pneumonia**, **hypoglycaemia**, **hypothermia**, likely **HIV infection**, and diarrhoea.

2. What are your immediate treatments (first hour)?

> - **Oxygen** by nasal prongs (target $SpO_2 \geq 90\%$)
> - **50 mL 10% glucose/sugar water by NG**, then **F-75 every 30 min for 2 h** (≈ 14 mL each), then 2-hourly 57 mL (11 mL/kg)
> - **Kangaroo care**, hat, blankets
> - **Ampicillin 260 mg (50 mg/kg) IV 6-hourly + gentamicin 39 mg (7.5 mg/kg) IV daily**; consider **ceftriaxone** if no improvement in 48 h
> - Assume dehydration from loose stools only if watery and recent change: **ReSoMal 5 mL/kg every 30 min**, watching for overload (pneumonia + SAM)

3. Which infections must you specifically consider in an HIV-exposed infant with SAM and pneumonia?

> **Pneumocystis jirovecii pneumonia (PCP)** (hypoxia, age 2–6 months+, HIV-exposed, not on co-trimoxazole) → **high-dose co-trimoxazole** and prednisolone per protocol; **TB** (contact = mother; CXR, Xpert on gastric aspirate/stool); bacterial pneumonia (*S. pneumoniae*, *S. aureus*, Gram-negatives); CMV; **oral/oesophageal candidiasis** (nystatin/fluconazole).

4. What investigations do you order?

> **HIV DNA PCR** (she is < 18 months; antibody test only shows exposure), CXR, **Xpert MTB/RIF Ultra** on gastric aspirate or stool, FBC/Hb, malaria RDT, blood culture, urine dipstick and culture, electrolytes, and the **mother's viral load**.

5. When will you start ART, and what else does she need for HIV?

> Start **ART as soon as possible after stabilisation** of metabolic complications (UCG 2023), typically within 1–2 weeks, once PCR is positive (start immediately on a positive result while sending confirmation). **Co-trimoxazole prophylaxis** (or treatment dose if PCP). Re-engage the **mother on ART** with adherence counselling and a viral load. Screen siblings.

6. What is her prognosis and how will you discuss it with the mother?

> SAM with HIV carries **high inpatient mortality** and slower recovery. Explain honestly but hopefully: the next 48 hours are the most dangerous; she needs **feeds every 2 hours day and night**, warmth, antibiotics and oxygen; with ART and nutritional rehabilitation many children recover well. Support the mother non-judgementally to restart her own ART.

Take-away: SAM with HIV and pneumonia: treat hypoglycaemia, hypothermia and hypoxia at once, give broad antibiotics, think PCP and TB, confirm HIV by DNA PCR, and start ART soon after stabilisation while re-engaging the mother in care.

Case 3: Growth faltering at a HC III immunisation clinic

At a HC III in Mukono, a 12-month-old boy, Daniel, is brought for his measles–rubella vaccine (overdue). The nurse notices his weight on the child health card has not changed since 8 months (7.4 kg). His mother says he is 'fine but small'.

1. What will you measure, and how do you interpret the card?

> **Weight 7.3 kg, length 72 cm** → WLZ about −2.4, LAZ about −1.7, **MUAC 12.1 cm**, **no oedema**. The card shows **growth faltering for 4 months** (flat line, now crossing below −2 WAZ). Classification: **MAM (moderate wasting)**; no danger signs; passes appetite test.

2. What history will you take?

> **Feeding**: breastfeeding? (yes, 4 times/day); complementary foods started at 5 months, **thin maize porridge twice a day**, family matooke sometimes, no animal-source food, shares a plate with siblings. **Illnesses**: 3 episodes of diarrhoea and malaria in 4 months. **HIV status** of mother (negative at ANC, not retested). **Deworming** and vitamin A: not received. **Social**: mother works in the garden all day; grandmother feeds.

3. What will you do today?

> - Give the **MR vaccine** today (and any other missed vaccines)
> - **Vitamin A 200,000 IU** (routine dose from 12 months; 100,000 IU at 6–11 months)
> - **Albendazole 200 mg** (12–23 months)
> - **Malaria RDT** if febrile; offer **HIV retest** to mother
> - **Counsel** (grandmother too): 3–4 meals + 1–2 snacks, ¾ cup (250 mL cup) per meal, **thick porridge with oil/groundnut paste**, add egg, fish, beans, dark green leaves, orange fruits; **own plate**, responsive feeding; breastfeed on demand; hand-washing
> - Consider **LNS/RUSF** if available (WHO 2023 high-risk: age < 24 months)
> - Return **immediately** if danger signs; **review in 2–4 weeks** (UCG: 30 days)

4. He returns after 1 month: MUAC 11.2 cm, weight 7.0 kg, appetite poor at the test, no oedema. What now?

> He now has **SAM** (MUAC < 115 mm) and **failed the appetite test** = **complicated SAM → refer for inpatient care**: give **sugar water**, keep warm, **first dose of ampicillin + gentamicin**, referral note with measurements. Ask about new illness (TB contact, cough, HIV).

5. What went wrong, and what system lesson is there?

> Faltering was noticed at **8 months** but no action was taken until he was 12 months: **growth monitoring without promotion**. Every flat or falling line should trigger assessment, counselling and early follow-up (and VHT home visits). Missing vaccines, vitamin A and deworming should be caught at every contact.

Take-away: Growth monitoring only saves lives when a flattening line triggers action: assess feeding and illness, counsel the actual feeder, catch up vaccines, vitamin A and deworming, and follow up before MAM becomes SAM.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.